

WORK EXPERIENCE

Thinking Machines Lab

New York City, NY

Member of Technical Staff, currently among the 40 longest tenured employees

Oct 2025 – current

- Developed multimodal pre-training and post-training data, and long horizon agentic post-training and evaluation towards mainline model training of [Interaction](#) and [Inkling Models](#).
- Co-developed pre-training and post-training evaluation infrastructure – core evaluation suites across verticals; reliable, fault-tolerant and throughput-optimized inference; and an interactive dashboard to visualize and compare model/grader rollouts, metrics and configurations.
- Skills: Python, PyTorch, Ray Core, Spark, Kafka, SGLang, Databases

Apple Foundation Models

New York City, NY

Machine Learning Researcher, LLMs

March 2023 – Oct 2025

- Large Language Model (LLM) research as part of the core model training team for Foundation Models (FM) that power Apple Intelligence and FM Developer API [\[blog\]](#)[\[technical report\]](#)
- Led efforts on pre-training and evaluation of language models, with a focus on web scale data, pretraining recipes and multilinguality, with multiple work streams that collectively engaged 25+ contributors
- Skills: Python, Jax, Apache Beam on Dataflow, AXLearn/Ajax LLM training frameworks

Meta AI (Facebook AI Research)

Menlo Park, CA; Pittsburgh, PA

Research Engineer, AI for Creativity

July 2019 – March 2023

- Lead Engineer for AI for Creativity supporting exploratory research in animating human characters with transformers and RL. Developed and open-sourced the [fairmotion](#) and [ScaDiver](#) libraries [\[12\]](#)
- Initiated and led cross-functional projects to apply research to production. Established collaborations with Habitat for Embodied AI research, Meta Horizon Worlds VR product, Sanzaru game studio, FAIR Paris/Menlo Park, and The Movement Lab at Stanford University
- Published research at ACM SIGGRAPH [\[7\]](#) [\[10\]](#) [\[11\]](#), NeurIPS [\[9\]](#), and SIGGRAPH Asia [\[8\]](#)
- Tech lead for multiple projects involving 20+ engineers in total at Meta AI. Mentored 9 Research and SWE interns, supported 20+ new team members and mentored 5 junior engineers in technical mentorship programs
- Skills: Python, PyTorch, TorchServe; RLlib for distributed RL training; PyBullet and Isaac Gym physics simulators; OpenGL, Blender, Unity for character/scene building, animation and rendering.

Research Engineer, Language and Translation Technologies

Jan 2017 – July 2019

- Co-developed and shipped the first neural machine translation (NMT) system to power translations for all user content on Facebook and Instagram (7B+ translations per day) [\[blog\]](#)[\[press\]](#)
- Tech lead for a team that modeled and deployed the first multilingual NMT systems to production [\[blog\]](#) [\[press\]](#)
- Managed “engineering excellence” projects towards optimized workflows for data processing, model training and evaluation, transitioning models to PyTorch, addressing reproducibility, determinism and safety in translations.
- Skills: Python, PyTorch; ONNX, C++ for production code; Hive, Spark for data processing

Software Engineering Intern, Language Technology

May 2016 – Aug 2016

- Developed neural network language models on the (newly) open sourced deep learning framework Caffe2.0

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

M.S., Language Technologies Institute, School of Computer Science

Aug 2015 – Dec 2016

Birla Institute of Technology and Science, Pilani

Pilani, India

B.S. in Computer Science

2010 – 2014

EARLY CAREER

LinkedIn

Software Engineer, Content Search

Bangalore, India
July 2014 – Aug 2015

- Developed the prefix query tagger library to predict autocompletions and tags for an incomplete query
- Co-implemented and deployed federated search and auto-complete search for several verticals on linkedin.com
- Actively involved in recruitment and onboarding of university grad hires and co-mentored 3 SWE interns
- Skills: Java, Lucene, Pig, Flex, Play Framework with JQuery and Bootstrap, JavaScript

Microsoft Research

Research Developer Intern, Applied Sciences

Bangalore, India
Jan 2014 – July 2014

- Fraud detection in Azure services and sentiment analysis of Azure users in social media
- Launched the Indian Elections Game web app, an interactive prediction market for the public to make predictions on the 2014 Indian Elections [\[blog\]](#)

Amazon

Machine Learning Intern

Bangalore, India
May 2013 – July 2013

- Adult content detection for Amazon Android AppStore, leading to a 30% load reduction for human reviewers
- Developed a system to cluster Amazon seller FAQs using Latent Dirichlet Allocation (LDA) and map customer queries to the clusters with logistic regression, SVMs, and supervised LDA

AWARDS AND HONORS

- ACM ICMI Ten-Year Technical Impact Award Runner-Up Award for the paper “Deep Multimodal Fusion for Persuasiveness Prediction” presented 5-9 October 2026 in Napoli, Italy
- Project Acyut: A team of 5 undergraduate students that built India’s first indigenously developed autonomous humanoid robot. I designed and implemented all aspects of the computer vision and robot behavior modules [\[press coverage\]](#)
 - 3rd Prize at RoboCup 2011 in Autonomous Humanoid Team Soccer
 - 2 silver medals at RoboGames 2011, the (then) world’s biggest open robotics event
- Placed 4th in ACM ICPC, Asia Regionals, Kharagpur site in December 2013
- Placed 1st in LinkedIn Bangalore HackDay 2015 and 2nd in Amazon Machine Learning HackEnd 2013

PUBLICATIONS

- [1] “Over-Searching in Search-Augmented Large Language Models” [\[pdf\]](#)
R. Xie, **D. Gopinath**, D. Qiu, D. Lin, H. Sun, S. Potdar, and B. Dhingra
19th Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2026.
- [2] “Interleaved Reasoning for Large Language Models via Reinforcement Learning” [\[pdf\]](#)
R. Xie, D. Qiu, **D. Gopinath**, D. Lin, Y. Sun, C. Wang, S. Potdar, B. Dhingra
NeurIPS Workshop on Multi-Turn Interactions in Large Language Models (MTI-LLM), 2025.
- [3] “Apple Intelligence Foundation Language Models” [\[pdf\]](#)
T Gunter, Z Wang, C Wang, R Pang, ..., **D. Gopinath**, Apple Inc.
arXiv preprint arXiv:2407.21075, 2024.
- [4] “Simulation and Retargeting of Complex Multi-Character Interactions” [\[pdf\]](#)
Y Zhang, **D. Gopinath**, Y Ye, J Hodgins, G Turk, J Won
ACM Special Interest Group on Computer Graphics (SIGGRAPH), 2023.
- [5] “CIRCLE: Capture In Rich Contextual Environments” [\[pdf\]](#)
JP Araújo, J Li, K Vetrivel, R Agarwal, J Wu, **D. Gopinath**, AW Clegg, K Liu
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.

- [6] “Motion In-betweening for Physically Simulated Characters” [\[pdf\]](#)
D. Gopinath, J. Won, and H. Joo
ACM Special Interest Group on Computer Graphics Asia (SIGGRAPH Asia), 2022.
- [7] “Physics-based Character Controllers using Conditional VAEs” [\[webpage\]](#)[\[pdf\]](#)
 J. Won, **D. Gopinath**, and J. Hodgins
ACM Special Interest Group on Computer Graphics (SIGGRAPH), 2022.
- [8] “Transformer Inertial Poser: Attention-based Real-time Human Motion Reconstruction from Sparse IMUs” [\[pdf\]](#)
 Y. Jiang, Y. Ye, **D. Gopinath**, J. Won, A.W. Winkler, and C.K. Liu
ACM Special Interest Group on Computer Graphics Asia (SIGGRAPH Asia), 2022.
- [9] “Leveraging Demonstrations with Latent Space Priors”
 J. Gehring, **D. Gopinath**, J. Won, G. Synnaeve, A. Krause, and N. Usunier
Conference on Neural Information Processing Systems (NeurIPS) (under review), 2022.
- [10] “Control Strategies for Physically Simulated Characters Performing Two-player Competitive Sports”.
[\[webpage\]](#)[\[pdf\]](#)
 J. Won, **D. Gopinath**, and J. Hodgins
ACM Special Interest Group on Computer Graphics (SIGGRAPH), 2021.
- [11] “A Scalable Approach to Control Diverse Behaviors for Physically Simulated Characters” [\[webpage\]](#)[\[pdf\]](#)
 J. Won, **D. Gopinath**, and J. Hodgins
ACM Special Interest Group on Computer Graphics (SIGGRAPH), 2020.
- [12] “fairmotion - Tools to load, process and visualize motion capture data” [\[code\]](#)
D. Gopinath and J. Won
GitHub, facebookresearch/fairmotion, 2020.
- [13] “Harnessing Indirect Training Data for End-to-End Automatic Speech Translation: Tricks of the Trade” [\[pdf\]](#)
 J. Pino, L. Puzon, J. Gu, X. Ma, A. D. McCarthy, and **D. Gopinath**
International Conference on Spoken Language Translation (IWSLT), 2019.
- [14] “Deep Multimodal Fusion for Persuasiveness Prediction” [\[pdf\]](#)
 B. Nojavanasghari*, **D. Gopinath***, J. Koushik*, T. Baltrušaitis, and L. P. Morency
ACM International Conference on Multimodal Interaction (ICMI), 2016.